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**Aim:** To study and implement Version Control using Git & GitHub.

**Theory:**

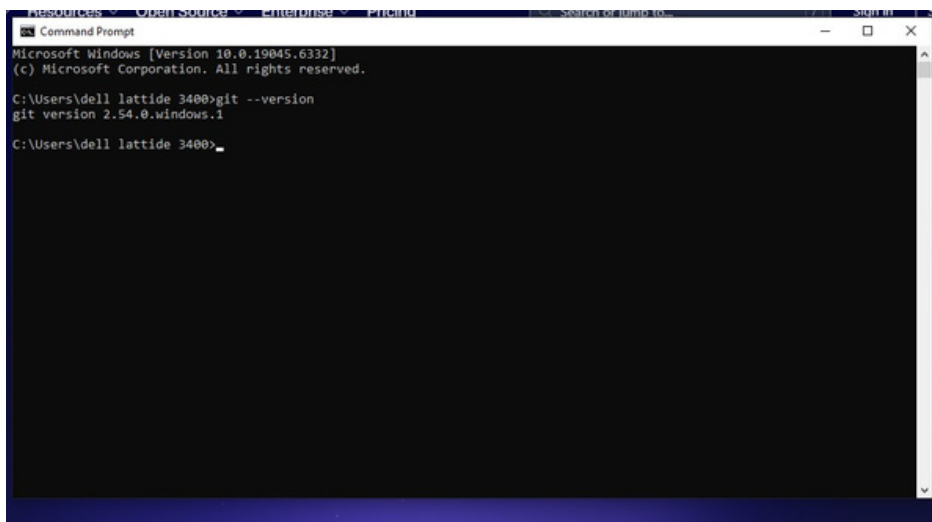
Git is a distributed version control system used to track changes in source code during software development. GitHub is a cloud-based platform used to host Git repositories online. It helps developers collaborate, manage code versions, and maintain project history efficiently.

**Step by Step Procedure:**

1. Installation of Git
2. Configure username & email
3. Create local repository
4. Create files
5. Add files to staging area
6. Commit changes
7. Create GitHub repository
8. Link local repository to GitHub
9. Push files to GitHub

**Screenshots:**

**Git Installation Verification**



```
Microsoft Windows [Version 10.0.19045.6332]
(c) Microsoft Corporation. All rights reserved.

C:\Users\dell latitude 3400>git --version
git version 2.54.0.windows.1

C:\Users\dell latitude 3400>
```

**Repository Creation**

New repository x + Ask Gemini

github.com/new

### Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository.](#)  
Required fields are marked with an asterisk (\*).

1 General

Owner \*  bodkhesakshi09-design / Repository name \*

⚠ Name cannot be blank

Great repository names are short and memorable. How about [ubiquitous-octo-guacamole](#)?

Description

0 / 350 characters

2 Configuration

Choose visibility \*

Choose who can see and commit to this repository

Add README ☐ Off

READMEs can be used as longer descriptions. [About READMEs](#)

## Commit History

github - Search x Commits - bodkhesakshi09-design x +

https://github.com/bodkhesakshi09-design/myproject/commits/main/

### Commits

main All users All time

Commits on May 5, 2026

|                                                                                                                                |          |         |                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------|----------|---------|------------------------------------------------------------------------------------------|
| Add files via upload                                                                                                           | Verified | 6681b95 |  <>   |
|  bodkhesakshi09-design authored 1 hour ago    |          |         |                                                                                          |
| Add files via upload                                                                                                           | Verified | 145483b |  <>   |
|  bodkhesakshi09-design authored 1 hour ago    |          |         |                                                                                          |
| Add files via upload                                                                                                           | Verified | d8356a  |  <> |
|  bodkhesakshi09-design authored 1 hour ago  |          |         |                                                                                          |
| Add files via upload                                                                                                           | Verified | b366887 |  <> |
|  bodkhesakshi09-design authored 2 hours ago |          |         |                                                                                          |
| Add files via upload                                                                                                           | Verified | 2a6401b |  <> |
|  bodkhesakshi09-design authored 2 hours ago |          |         |                                                                                          |
| Add files via upload                                                                                                           | Verified | 9c4c109 |  <> |
|  bodkhesakshi09-design authored 2 hours ago |          |         |                                                                                          |
| Add files via upload                                                                                                           | Verified | fe69f5  |  <> |
|  bodkhesakshi09-design authored 3 hours ago |          |         |                                                                                          |

## GitHub Repository Page

Repositories x + Ask Gemini

github.com/repos?q=owner%3A%40me

Repositories

My contributions

My repositories

My forks

Admin access

Views

No saved views

### My repositories

owner:@me

2 repositories

Relevance

bodkhesakshi09-design/My-first-project Private

This is my first GitHub project

0 0 0 0 Updated on Nov 21, 2025

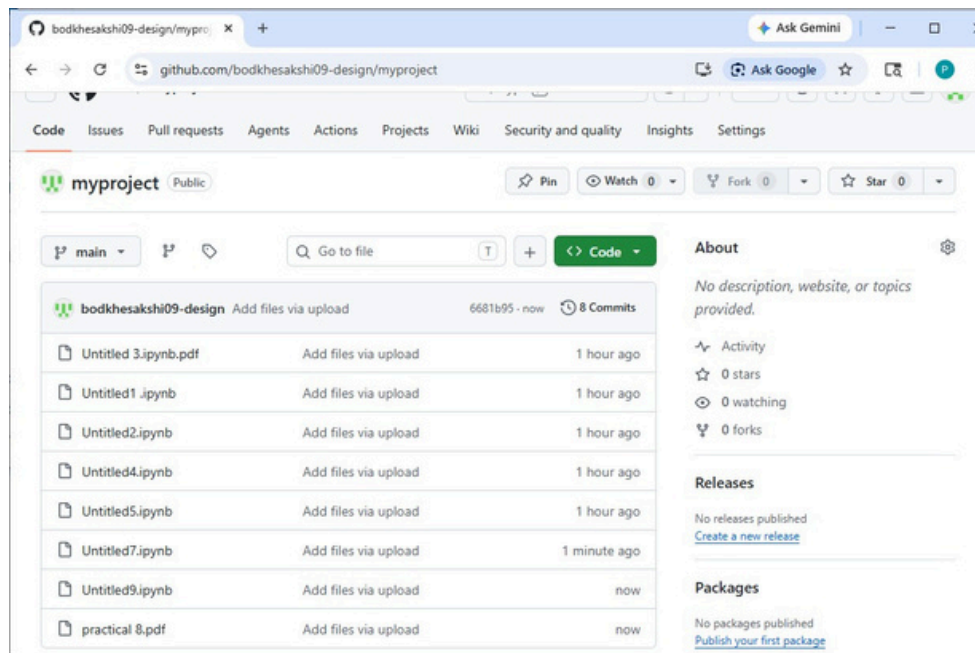
bodkhesakshi09-design/myproject Public

Jupyter Notebook 0 0 0 0 Updated 7 minutes ago

**ProTip!** Using a comma within certain filters will aggregate the results. For example: language:Go,Ruby will show all repositories that have either Go or Ruby as their primary language.

Type here to search

## Files on GitHub Repository



## Questions & Answers:

### 1. What is the difference between Git and GitHub?

Git is a distributed version control system used to track changes in source code, documents, and project files. It helps developers manage different versions of a project and allows multiple people to work together efficiently. Git works on a local computer and keeps the history of all changes made in the project.

GitHub is a cloud-based platform that hosts Git repositories online. It provides features like online storage, collaboration, project management, sharing repositories, and backup. GitHub allows developers to upload their local Git repositories to the internet so that team members can access and contribute to the project from anywhere.

In simple words, Git is a tool for version control, while GitHub is a website or platform used to store and manage Git repositories online.

2. Explain the difference between git add and git commit.

git add

The git add command is used to add files or changes to the staging area. Before saving changes permanently, Git first collects the modified files in the staging area. This command prepares the files for commit.

Example:

git add .

This command adds all modified files to the staging area.

git commit

The git commit command is used to permanently save the staged changes in the local repository. Every commit creates a snapshot of the project at a particular time.

Example:

git commit -m "First commit"

This command saves the changes with a message called commit message.

Difference:

- git add prepares files for saving.
- git commit permanently stores the changes in Git history.

3. What is the staging area in Git?

The staging area is an intermediate area between the working directory and the Git repository. It is used to prepare and organize files before committing them.

When a file is modified, Git does not directly save the changes permanently. First, the file is added to the staging area using the git add command. After reviewing the changes, the user commits them using git commit.

The staging area gives better control over which changes should be included in the next commit. It helps developers manage project updates more efficiently.

4. What happens if you modify a file after committing it?

When a file is modified after committing, Git identifies it as a changed or modified file. The previous commit remains unchanged, but the new modifications are not automatically saved in Git history.

To save the updated version, the user must:

- Add the modified file again using git add
- Create another commit using git commit

This process allows Git to maintain a complete history of all changes made to the project over time.

Developers can also compare old and new versions of files whenever required.

5. Difference between git push and git pull.

git push

The git push command is used to upload local repository changes to a remote repository such as GitHub. It sends commits from the local system to the online repository.

Example:

Bash

git push origin master

git pull

The git pull command is used to download the latest updates from the remote repository to the local system. It helps synchronize the local project with GitHub.

Example:

Bash

git pull origin master

Difference:

git push uploads local changes to GitHub.

git pull downloads remote changes to the local repository.

6. What is Version Control? Differentiate between Centralized and Distributed VCS.

Version Control is a system used to manage and track changes in files, documents, and source code over time. It helps developers maintain project history, recover old versions, and support teamwork in software development.

Advantages of Version Control:

- Tracks file changes

- Maintains project history

- Supports collaboration

- Helps recover previous versions

- Reduces chances of data loss

Centralized Version Control System (CVCS)

In a centralized version control system, there is one central server that stores all project files and version history. Users connect to the central server to access the project.

Features:

- Single central repository

- Requires internet/network connection

- Easier administration

- Risk of data loss if server fails

Example:

SVN (Subversion)

Distributed Version Control System (DVCS)

In a distributed version control system, every user has a complete copy of the repository along with its history. Users can work offline and later synchronize changes with the remote repository.

Features:

- Complete repository available to every user

- Faster operations

- Supports offline work

- Better backup and security

Example:

Git

Difference:

CVCS uses one central repository.

DVCS provides a complete repository copy to every user.

Git is an example of Distributed VCS.